

**Khan Institute of Computer Science & Information
Technology**

Open-Ended Lab Project
**Artisan Core: A Multidisciplinary Art
Intelligence System**

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1 PROJECT INTRODUCTION

Artisan Core is a professional Decision Support System (DSS) developed using the ASP.NET Core MVC framework. The primary goal of this project is to provide automated, intelligent roadmaps for creators across various artistic domains such as Fine Arts, Web Development, Fashion Design, and Handmade Crafts (Crochet).

1.1 Problem Statement

Artists often struggle with project management and technical roadmapping. Artisan Core solves this by providing a structured 3-phase execution plan based on the specific art category entered by the user.

2 SYSTEM ARCHITECTURE & DESIGN

The system follows the Model-View-Controller (MVC) architectural pattern, ensuring a clean separation of concerns between the user interface and the underlying logic.

2.1 Visual Programming Implementation

- **Frontend (The View):** Developed using Razor Pages and Bootstrap 5 with a custom "Glassmorphism" aesthetic.
- **Logic (The Controller):** Implemented keyword-based pattern matching in C# to identify art categories and generate contextual data.
- **Data Binding:** Seamlessly transfers user input from the search interface to the analysis engine and back to the visual components.

3 KEY FEATURES & FUNCTIONALITIES

- **Dynamic Image Switching:** The UI automatically updates high-quality visual references using the Unsplash API integration based on the detected art type.
- **Phase-Wise Roadmap:** Generates specific instructions for Phase 1 (Concept), Phase 2 (Execution), and Phase 3 (Precision).

- **Aesthetic User Experience:** Features a modern dark theme with neon accents and responsive layouts for professional-grade presentation.

4 DEPLOYMENT AND CONTAINERIZATION

To ensure the application runs consistently across different environments, the project includes a specialized Docker configuration.

4.1 Dockerfile Configuration

The application utilizes a multi-stage Docker build process:

- **Build Stage:** Uses the .NET 10 SDK to restore and publish the application.
- **Run Stage:** Uses the ASP.NET 10 runtime to host the containerized application efficiently on port 8080.

5 CONCLUSION

Artisan Core demonstrates the power of visual programming and modern web frameworks in solving real-world creative challenges. By integrating intelligent logic with a premium user interface, the system provides a scalable solution for project management in the arts industry.